

Section A – Concept Application

Question 1: Selecting Compatible Computer Components

Answer:

When selecting components for a new office workstation, I would first choose a motherboard that supports the required CPU socket and chipset. Then I would select a compatible CPU based on the motherboard's socket and supported CPU list.

Next, I would choose RAM that matches the motherboard's supported memory type, speed and maximum capacity. For storage, I would check whether the motherboard supports the required SATA or M.2/NVMe interface.

To confirm compatibility, I would check the motherboard manufacturer's specifications and CPU support list, RAM compatibility, available storage interfaces, and the required power connections. I would also verify that the PSU can provide enough power for all components. Finally, I would check the system specifications before assembly to make sure all selected components are compatible.

Question 2: HDD, SSD and NVMe Storage Comparison

Answer:

HDD (Hard Disk Drive): HDDs store data on spinning magnetic disks. They usually offer large storage capacity at a lower cost, but they are slower and have moving mechanical parts.

SSD (Solid State Drive): SSDs use flash memory and have no moving parts. They provide faster boot times and file access than HDDs and are more resistant to physical movement.

NVMe SSD: NVMe SSDs are a type of SSD that use the PCIe interface and NVMe protocol. They generally provide much higher data-transfer performance and lower latency than traditional SATA SSDs.

Recommendation: For modern office workstations, I would recommend an **SSD, preferably an NVMe SSD** if the budget and motherboard support it. It provides fast application loading, quick boot times and responsive file access, which improves the overall productivity experience.

Question 3: Role of GPU and Cooling System

Answer:

The **GPU** is responsible for processing and rendering graphics and visual output. For a workstation running mildly graphics-intensive design software, the GPU helps handle graphics calculations and provides smooth visual performance.

The **cooling system** removes heat produced by the CPU, GPU, and other components during operation. Good cooling helps prevent overheating and allows the system to maintain stable performance during several hours of continuous use.

Conclusion: The GPU provides the required graphics performance, while the cooling system helps keep the components within safe operating temperatures and maintains system stability under sustained load.

Question 4: Importance of Hardware Compatibility

Answer:

Hardware compatibility is important because all computer components must work together correctly. The motherboard must support the CPU socket and chipset, the RAM type and speed, and the storage interface.

If incompatible components are used, the computer may fail to boot, the component may not be detected, or the system may become unstable. For example, an unsupported CPU may not work with the motherboard, and incompatible RAM may cause boot or stability problems.

Before assembling the computer, I would check the motherboard manufacturer's specifications and compatibility information for the CPU, RAM, storage, GPU, and PSU.

Question 5: Selecting and Connecting the Power Supply Unit (PSU)

Answer:

I would choose a PSU with enough wattage to safely power the motherboard, CPU, GPU, storage devices, and other components. I would also consider the PSU's efficiency rating and make sure it has the required power connectors for the build.

Before powering on the system, I would make sure the PSU is switched off, and the AC power cable is disconnected while making connections. I would check that the motherboard 24-pin ATX connector and CPU power connector are firmly connected.

I would connect the required PCIe power cable to the GPU and the appropriate SATA power cables to SATA storage devices. I would check that all connectors are fully seated and that cables are not damaged or loose.

Finally, I would verify all connections, make sure there are no loose screws or objects inside the case, and only then connect the AC power cable and switch on the PSU.

Question 6: Confirming Successful POST

answer

After assembling the workstation, I would power on the system and check whether it completes the Power-On Self-Test (POST) successfully. I would check that the display appears, the system starts normally, and there are no unusual beep codes or POST error messages.

I would also enter the BIOS/UEFI setup and confirm that important components such as the CPU, RAM, and storage device are detected correctly.

If the system does not pass POST, I would switch off the PC and check the basic hardware connections first. I would inspect the RAM seating, CPU power connection, motherboard power connection, GPU/display connection, and storage connections. I would then restart the system and check the POST result again.

Section B

Task 1: Component Identification & Functions

No.	Component	Function	Make/Model or Key Specification
1	Motherboard	Connects and allows communication between the main computer components.	Zebronics H110
2	CPU	Processes instructions and performs the main calculations of the computer.	Intel Core i5-6500
3	RAM	Temporarily stores data and instructions currently being used by the system.	16 GB DDR4
4	GPU	Processes graphics and provides visual output to the display.	No Dedicated GPU / Integrated Graphics
5	Storage Device	Store the operating system, applications and user files permanently.	256 GB NVMe SSD + 4 TB HDD
6	Cooling System	Removes heat from the CPU and helps maintain stable operation.	CPU Fan/Cooler

Practical Identification

The components were identified based on the available system specifications and written documentation.

Task 2: Motherboard Ports & Socket Identification

I identified the main physical ports and slots on the motherboard and noted their purpose. I also identified the CPU socket type and the compatible CPU family.

Motherboard Ports & Slots

Port / Slot	Purpose
SATA Ports	Used to connect SATA storage devices such as HDDs and SATA SSDs.
24-Pin ATX Power Connector	Provides main power to the motherboard.
CPU Power Connector	Supplies power to the CPU.
USB Headers	Connect the front-panel USB ports to the motherboard.
PCIe Slot	Used for expansion cards such as graphics cards and other PCIe devices.
M.2 Slot	Used for compatible M.2 storage devices such as NVMe SSDs.
RAM Slots	Used to install system memory modules.

CPU Socket: LGA 1151

**Compatible CPU Family: 6th and 7th
Generation Intel Core processors**

Task 3: RAM & Storage Installation

Written Documentation / Simulated Procedure

1. ESD Precautions

I would shut down the PC, disconnect the power cable, and use an anti-static wrist strap or other proper ESD precautions before handling internal components.

2. Remove the RAM

I would release the RAM slot clips and carefully remove the RAM module from the motherboard.

3. Reinstall the RAM

I would align the notch on the RAM with the slot and press it down evenly until the retaining clips lock into place.

4. Remove the Storage Device

For a SATA drive, I would disconnect the SATA data and power cables. For an NVMe drive, I would remove the retaining screw and carefully remove the M.2 drive.

5. Reinstall the Storage Device

I would reconnect the SATA cables, or reinstall the NVMe drive and secure it with its screw.

6. Confirm Detection

After reassembly, I would start the PC and enter BIOS/UEFI to check whether the RAM and storage device are detected correctly.

Expected Result:

The RAM should show the installed memory capacity, and the storage device should appear in the BIOS/UEFI storage information.

Task 4: Full Desktop PC Assembly

Desktop PC Assembly Procedure

1. ESD and Safety Preparation

I would switch off and unplug the system and take proper ESD precautions before handling the components.

2. Install the CPU

I would carefully place the compatible CPU into the motherboard's CPU socket and secure it correctly.

3. Install the CPU Cooling System

I would install the CPU cooler correctly to provide proper cooling during operation.

4. Install the RAM

I would align the RAM module with the motherboard slot and press it down until it is properly seated.

5. Install the Storage Device

I would install the storage device using its

supported connection, such as SATA or M.2/NVMe.

6. Install the Motherboard and GPU

I would mount the motherboard securely inside the case. If a dedicated GPU is provided, I would install it into the appropriate PCIe slot.

7. Connect the PSU

I would connect the required motherboard, CPU, GPU and storage power connectors from the PSU.

8. Check All Connections

I would verify that all components and cables are properly connected and that there are no loose objects inside the case.

9. First Boot and POST Check

I would power on the system and check whether it completes POST successfully. I would also check BIOS/UEFI to confirm that the CPU, RAM and storage are detected.

Conclusion

Following the correct assembly sequence and ESD precautions helps prevent component damage and installation errors. A successful POST and correct hardware detection confirm that the basic assembly is working correctly.

Section C – Mini Capstone Project

Hardware Components to Check First

When a newly assembled PC fails to boot, I would first check the **PSU, RAM, and motherboard**.

1. PSU: I would check the PSU power connection and make sure the motherboard and CPU power connectors are properly connected. A power problem can prevent the system from starting.

2. RAM: I would check whether the RAM is correctly seated in the motherboard slot. Incorrectly installed RAM can cause POST or boot problems.

3. Motherboard: I would check the motherboard for loose connections, incorrect installation, or compatibility issues. The motherboard is responsible for connecting and communicating with the main hardware components.

Checking these components first helps identify common installation and power-related causes of a boot failure.

2. Motherboard–CPU Compatibility

Motherboard–CPU compatibility is important because the motherboard must support the CPU socket, chipset, and processor family. If the CPU is not compatible with the motherboard, the computer may fail to boot or may not work correctly.

In the system I inspected, the **Zebronics H110 motherboard** uses an **LGA 1151 socket**, and the **Intel Core i5-6500** is a compatible processor for this platform. Therefore, the motherboard and CPU can work together. I would confirm compatibility by checking the motherboard specifications and CPU support information before installation.

3. RAM and Storage Installation

Verification

I would verify that the RAM module is correctly seated in the motherboard slot and that the storage device is properly connected.

After starting the PC, I would enter the BIOS/UEFI and check whether the installed RAM capacity is detected correctly. I would also check whether the storage device appears in the BIOS/UEFI storage information.

A successful POST should detect the RAM and basic storage hardware without showing related hardware errors. If either component is not detected, I would power off the system and check the connections and seating again.

4. Hardware Installation Checklist

1. Check Compatibility: Verify that the motherboard, CPU, RAM, storage, GPU, and PSU are compatible before installation.

2. Follow ESD and Safety Precautions:

Power off and unplug the system and use proper ESD precautions while handling components.

3. Verify Connections: Check that RAM, storage, power connectors, and other components are properly seated and connected before the first boot.

4. Perform a POST Check: Start the PC and confirm that it completes POST and that the main hardware components are detected in BIOS/UEFI.

5. SSD vs HDD for an Office Workstation

For a modern office workstation, I would recommend an **SSD** instead of an HDD. An SSD has no moving mechanical parts and provides faster boot times, application loading, and file access.

An HDD can provide a larger amount of storage at a lower cost, but it is generally slower because it uses moving mechanical parts.

For employees who mainly use office and productivity applications, an SSD provides better overall system responsiveness and productivity. If large amounts of low-cost storage are required, an HDD can be used as additional storage.

Section D – AI-Augmented Learning

Step 1: Build with AI

AI Prompt Used

Create a clear step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Put the steps in the correct priority order. Cover the motherboard, RAM, storage, and PSU as possible failure points. Include basic safety and ESD precautions. Apply the checklist to the ABC Solutions workstation boot-failure scenario from Section C and indicate which step could have identified the reported issue.

AI Original Troubleshooting Checklist

1.Check the PSU

- Make sure the power cable is connected and the PSU switch is ON.
- Check that the 24-pin motherboard power connector and CPU power connector are properly connected.

2.Check the Motherboard

- Check for loose connections or incorrect installation.
- Look for any visible damage or motherboard error indicators.

3.Check the RAM

- Make sure the RAM is fully seated in the correct slot.
- If the PC still does not boot, reseal the RAM and try again.

4.Check the Storage Device

- Check that the SATA or NVMe storage device is properly installed.
 - Confirm that it is detected in BIOS/UEFI.

5.Check POST/BIOS

- Power on the PC and check whether it completes POST.
 - Enter BIOS/UEFI and verify that the CPU, RAM and storage are detected.

6.Apply to ABC Solutions

- In the ABC Solutions scenario, these checks would help identify whether incorrect installation or hardware compatibility was responsible for the boot failure.

Step 2: Test & Debug (Without AI)

Issue Found:

The original AI checklist did not clearly put ESD and safety precautions as the first step before handling internal components.

Corrected Checklist

1.ESD and Safety Precautions – Shut down and unplug the PC and take proper ESD precautions.

2.Check the PSU – Check the power cable, PSU switch, 24-pin connector and CPU power connector.

3.Check the Motherboard – Check connections and installation.

4.Check the RAM – Make sure the RAM is properly seated.

5.Check POST/BIOS – Check POST and verify CPU/RAM detection.

6.Check Storage – Verify the SATA/NVMe connection and BIOS detection.

What I Changed and Why

I added ESD and safety precautions as the first step.

The original checklist did not clearly include this safety step.

This makes the troubleshooting process safer and helps prevent electrostatic damage.

I also made the physical hardware checks happen before BIOS/storage verification.

